Mahdi Kallel

PhD candidate in reinforcement learning

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Profile

PhD candidate in reinforcement learning at the University of Würzburg, advised by Prof. Carlo D'Eramo, graduating April 2027. I study how neural networks learn, especially in reinforcement learning, and how to turn that understanding into better training methods. Recently, I have been working on memory and plasticity in continual learning and on how learning dynamics differ between reinforcement and supervised learning. Earlier in my PhD, I worked on policy search and policy optimization.

Publications

2026 **Mahdi Kallel**, Nour Jamoussi, Johannes Tölle, Marios Kountouris, Carlo D'Eramo. **The Spectral Anatomy of Plasticity Loss**. *AXIOM Workshop at NeurIPS 2026*.
Uses random matrix theory to show that learned structure persists as plasticity declines, while later parameter updates increasingly concentrate in existing spectral directions.

2026 **Mahdi Kallel**, Johannes Tölle, Ahmed Hendawy, Carlo D'Eramo. **Do Not Imitate, Reinforce: Iterative Classification via Belief Refinement**. *Transactions on Machine Learning Research (TMLR), 2026*.
Trains classifiers to refine their predictions through reinforcement learning, improving the balance between accuracy and calibration compared with cross-entropy. The critic estimates remaining improvement and decides when to stop.

2026 **Mahdi Kallel**, Jose-Luis Holgado-Alvarez, Samuele Tosatto, Carlo D'Eramo. **PPO+: Enhancing Proximal Policy Optimization**. *Reinforcement Learning Journal (RLC 2026)*.
Shows that the critic is a bottleneck in PPO, and that adding off-policy data for critic training improves on-policy gradient estimates and performance.

2026 Ahmed Hendawy, Henrik Metternich, Théo Vincent, **Mahdi Kallel**, Jan Peters, Carlo D'Eramo. **Use the Online Network If You Can: Towards Fast and Stable Reinforcement Learning**. *ICLR 2026*.
Reduces the learning delay of target networks by taking the minimum of online and target value estimates, enabling faster learning while controlling overestimation.

2024 **Mahdi Kallel**, Debabrota Basu, Riad Akrour, Carlo D'Eramo. **Augmented Bayesian Policy Search**. *ICLR 2024*.
Introduces a Gaussian process prior that guides policy search with the policy gradient, enabling Bayesian optimization to scale to high-dimensional control problems.

2024 Oliver Järnefelt, **Mahdi Kallel**, Carlo D'Eramo. **Cyclicity-Regularized Coordination Graphs**. *RLC 2024*.
Shows that accounting for cycles when sparsifying coordination graphs improves joint action selection and performance in cooperative multi-agent reinforcement learning.

Research Experience

2022 **Research Intern**, *INRIA Lille, SCOOL Research Group*, France
- Started research on Bayesian policy search with deterministic policies, leading to Augmented Bayesian Policy Search (ICLR 2024).

2020 **Research Intern**, *EURECOM*, Biot, France
- Compared centralized and decentralized approaches to mobile traffic prediction and load balancing.

Industry Experience

2021 **Machine Learning Engineer**, *Mcovision (independent)*, Paris, France
- Built a classifier of echocardiography videos for cardiac amyloidosis and other rare heart diseases with Hôpital Cochin, from data cleaning to training.

2021 **Research Intern**, *SustainEcho*, Paris, France
- Explored few-shot and hierarchical learning methods for classification with little data.

Education

2023–2027 **PhD, Reinforcement Learning**, *University of Würzburg*, Germany
Expected April 2027. Supervisor: Prof. Carlo D'Eramo.
Teaching: Introduction to Machine Learning; Introduction to Reinforcement Learning.

2021–2022 **MSc, Optimization**, *Université Paris-Saclay*, France
Coursework: Convex analysis, dynamic programming, stochastic optimization and approximation.

2020–2021 **MSc, Mathematics, Vision, Learning (MVA)**, *ENS Paris-Saclay*, France
Coursework: Reinforcement learning, convex optimization, kernel methods, advanced deep learning.

2018–2020 **BSc, Data Science**, *Télécom Paris / EURECOM*, France

Supervision, Talks and Service

Supervision Johannes Tölle, MSc thesis, University of Würzburg (2026). The thesis received the VDE Bayern Award 2026, and Johannes co-authored two of the papers above.

Talks *On Better Priors for Policy Search*. Invited talk and lab visit, Improbable AI Lab (Pulkit Agrawal), MIT, August 2024.

Reviewer NeurIPS (top reviewer, 2026), ICLR, ICML, RLC

Technical Skills

Programming Python, JAX, PyTorch, NumPy, Julia, C

Languages Arabic (native), English (fluent), French (fluent), German (basic)